

Tribhuvan University
Faculty of Humanities & Social Sciences
OFFICE OF THE DEAN
Applied Economics
BCA — Semester 6 — 2024 — Model Answer Sheet
Subject Code: CAEC353

Note: This is a model answer sheet for study purposes. Answers are based on standard curriculum topics.

Group B

Attempt any SIX questions. [6×5=30]

2. Define economics. Distinguish between microeconomics and macroeconomics.

Answer: **Economics** is the social science that studies how individuals, businesses, governments, and societies make choices about allocating limited resources to satisfy unlimited wants. **Microeconomics:** Studies individual units (consumers, firms, industries). Topics: demand, supply, pricing, market structures. **Macroeconomics:** Studies economy as a whole. Topics: GDP, inflation, unemployment, monetary/fiscal policy.

3. Explain the law of demand and law of supply with diagrams.

Answer: **Law of Demand:** Other things equal, as price falls, quantity demanded rises (inverse relationship). Downward sloping demand curve. **Exceptions:** Giffen goods, Veblen goods. **Law of Supply:** Other things equal, as price rises, quantity supplied rises (direct relationship). Upward sloping supply curve. **Market Equilibrium:** Where demand and supply curves intersect. At this point, quantity demanded equals quantity supplied.

4. What is elasticity of demand? Explain the types of price elasticity of demand.

Answer: **Elasticity of Demand:** Measures responsiveness of quantity demanded to price changes. Formula: (% change in Qd) / (% change in P). **Types:** **Perfectly Elastic ($E_d = \infty$):** Any price increase causes demand to drop to zero. **Elastic ($E_d > 1$):** Luxury goods. **Unitary Elastic ($E_d = 1$):** Proportionate change. **Inelastic ($E_d < 1$):** Necessities. **Perfectly Inelastic ($E_d = 0$):** Essential medicines.

5. Describe the law of diminishing marginal returns with an example.

Answer: **Law of Diminishing Marginal Returns:** When increasing amounts of a variable input are added to fixed inputs, the marginal product eventually declines. **Example:** A factory with fixed machinery adds more workers. Initially, output increases rapidly. But after a point, each additional worker contributes less because they must share equipment. Eventually, overcrowding reduces total output. **Stages:** Increasing returns, diminishing returns, negative returns.

6. Differentiate between perfect competition and monopoly market structures.

Answer: **Perfect Competition:** Many sellers, homogeneous products, free entry/exit, perfect information, price takers. Examples: Agricultural markets. **Monopoly:** Single seller, no close substitutes, high barriers to entry, price maker. Examples: Utility companies, patents. **Comparison:** Perfect competition maximizes consumer

surplus and achieves allocative efficiency. Monopoly creates deadweight loss but may benefit from economies of scale and innovation incentives.

7. What is national income? Explain the methods of measuring national income.

Answer: National Income: Total value of goods and services produced by a country in a year. **Methods:**

Income Method: Sum of all incomes (wages, rent, interest, profit). **Expenditure Method:** Sum of all expenditures ($C + I + G + (X-M)$). **Value Added Method:** Sum of value added at each production stage.

Concepts: GDP (domestic), GNP (national), NNP (net), NI (national income at factor cost), PCI (per capita income).

8. Define inflation. Explain the causes and effects of inflation on the economy.

Answer: Inflation: Sustained increase in general price level over time. Measured by CPI, WPI, GDP deflator.

Causes: Demand-pull: Excess demand. **Cost-push:** Rising input costs. **Monetary:** Excess money supply.

Effects: Reduces purchasing power, hurts savers, benefits debtors, creates uncertainty, can lead to wage-price spirals, distorts resource allocation. **Control:** Monetary policy (interest rates), fiscal policy (taxes, spending), supply-side measures.

Group C

Attempt any TWO questions. [$2 \times 10 = 20$]

9. Explain the theory of consumer behavior using indifference curve analysis. Derive the demand curve from it.

Answer: Indifference Curve: Shows combinations of two goods giving equal satisfaction. **Properties:**

Downward sloping, convex to origin, never intersect. **Budget Line:** Shows affordable combinations given income and prices. **Consumer Equilibrium:** Where budget line is tangent to highest indifference curve ($MRS =$

price ratio). **Derivation of Demand Curve:** When price of good X falls, budget line pivots outward. New equilibrium on higher indifference curve with more X consumed. Plotting price-quantity pairs at different prices traces the demand curve. **Substitution Effect:** Switch to cheaper good. **Income Effect:** Increased purchasing power.

10. What is monetary policy? Discuss the tools of monetary policy used by the central bank to control inflation and deflation.

Answer: Monetary Policy: Central bank actions to control money supply and interest rates. **Tools:** 1) **Bank Rate:** Rate at which central bank lends to commercial banks. Increasing rate reduces money supply. 2) **Open Market Operations:** Buying/selling government securities. Selling reduces money supply. 3) **Cash Reserve Ratio (CRR):** Percentage of deposits banks must hold with central bank. Increasing CRR reduces lending capacity. 4) **Statutory Liquidity Ratio (SLR):** Percentage of deposits in liquid assets. 5) **Repo Rate:** Short-term lending rate. **Inflation Control:** Tight policy (higher rates, sell securities). **Deflation Control:** Expansionary policy (lower rates, buy securities).

11. Explain cost-output relationship in the short run and long run. Draw appropriate cost curves and explain their shapes.

Answer: Short Run: Some inputs fixed. **Costs:** **TFC:** Constant (horizontal line). **TVC:** Increases with output, initially at decreasing rate then increasing rate. **TC:** $TFC + TVC$. **MC:** U-shaped due to diminishing returns.

AVC, ATC: U-shaped. MC intersects AVC and ATC at their minimum points. **Long Run:** All inputs variable.

LAC: U-shaped due to economies and diseconomies of scale. **LMC:** Intersects LAC at minimum. **Economies of Scale:** Cost advantages from large-scale production. **Diseconomies:** Management difficulties at very large scale. **Minimum Efficient Scale:** Lowest point on LAC.

These model answers are prepared for educational purposes. Refer to textbooks for comprehensive study.